A Component-Level Evaluation of Web Interface Vulnerability Under Simulated Colour Vision Deficiencies

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***Abstract*—Abstract goes here**

***Index Terms*—Keywords go Here**

I. Introduction

Web accessibility ensures that websites and online systems align with the POUR principles: Perceivable, Operable, Understandable, and Robust, supporting users with visual, auditory, motor, and cognitive differences [1]. As the web increasingly functions as a gateway to education, employment, and services, inaccessible design risks excluding a significant portion of the population [1], [2].

An estimated 300 million individuals worldwide experience some form of Colour Vision Deficiency (CVD), including around 1 in 12 men and 1 in 250 women [3]. Because colour is widely used to communicate states such as alerts, navigation cues, and data visualisations, users with CVD may experience reduced clarity when interacting with colour-dependent interfaces [3], [4].

While prior studies provide insight into colour perception and interface usability, many evaluate interfaces as a whole or focus on predefined colour schemes [4], [5]. Limited research has examined how individual UI component categories behave under simulated CVD conditions across multiple websites, revealing a gap in structured component-level contrast analysis [2], [6].

A. Positioning

As shown in Figure 1, the Research Onion represents the layered methodological decisions guiding this study. This research adopts a philosophy of realism, viewing contrast ratios as measurable phenomena evaluated against WCAG thresholds [1]. A deductive, mono-method quantitative approach is used [2], following an experimental design in which websites are analysed under normal and simulated CVD conditions. The independent variable is the simulated vision condition; the dependent variable is the contrast performance of UI components. The time horizon is cross-sectional.

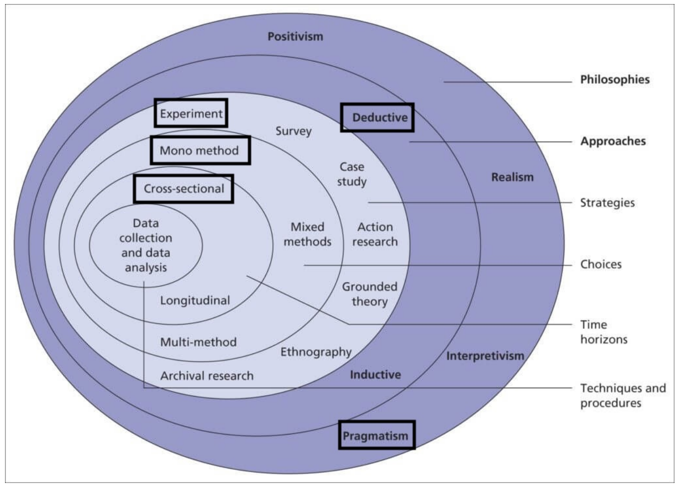

Fig. 1: Research onion illustrating philosophical and methodological positioning

B. Hypothesis

Simulated CVD conditions will reveal measurable differences in foreground–background contrast performance across UI component categories, indicating patterns of accessibility vulnerability.

C. Research Aim and Questions

This study aims to evaluate how different UI component categories perform in terms of foreground–background contrast under simulated CVD conditions [3], guided by the following questions:

- To what extent do UI components maintain acceptable contrast thresholds under simulated CVD conditions?
- How does contrast performance vary across component categories under simulated CVD conditions?
- Which UI components demonstrate the highest levels of contrast vulnerability across multiple websites?

II. Literature Review

A. Accessibility Evaluation Approaches

Web accessibility evaluation methods are broadly classified into automated tools, expert inspection, and user testing [6]. Automated tools efficiently detect structural issues such as missing alternative text or insufficient contrast, but their rule-based nature limits contextual understanding [6]. Expert evaluation provides deeper analysis but lacks scalability and introduces subjectivity [2], [6]. User testing captures practical usability barriers most directly, but is resource-intensive and difficult to generalise [6]. Hybrid approaches improve coverage but inherit the cost and complexity of each method [6].

B. Evaluation Metrics

Alsaeedi proposes quantitative metrics including the Coverage Error Ratio (CER) and Web Accessibility Accuracy (WAA) to compare tools and webpages [6]. However, these metrics rely on automated tool outputs, which are prone to inconsistent reporting and undetected violations, limiting their reliability as standalone measures of accessibility quality [2], [6].

C. CVD Simulation and Colour Contrast

Simulation-based methods apply computational models to replicate colour vision impairment, enabling scalable and controlled evaluation without requiring participants with CVD [3]. Sajek et al. use perceptual models such as CIELAB to quantify colour differences under CVD conditions [4], while Troiano et al. focus on generating accessible colour palettes through optimisation [5]. Both studies, however, operate at the level of global colour schemes rather than individual interface components [4], [5].

D. Research Gap

Existing research evaluates accessibility predominantly at the page level, with limited attention to individual component categories [2], [6]. CVD research similarly focuses on colour schemes rather than component-level behaviour [3], [4]. This creates a need for structured evaluation that combines CVD simulation with component-level contrast analysis across multiple websites [7]. Figure 2 maps these relationships and highlights the gap that motivates this study.

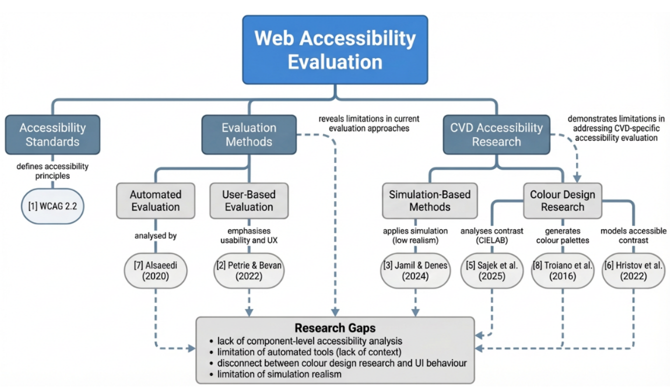

Fig. 2: Literature Map and Research Gaps

III. Research Methodology

A. Approach

This study follows a positivist, quantitative, and deductive methodology [2]. Accessibility is measured using contrast ratios, pass/fail outcomes, and failure rates aligned with WCAG 2.2 [1]. Rather than developing new theoretical frameworks, the study tests how well established guidelines perform at a component level under simulated CVD conditions [3], [4].

B. Experimental Design

A dataset of 20–30 publicly accessible websites is compiled across informational, commercial, and service-based domains. Websites requiring authentication are excluded to ensure a consistent and reproducible evaluation environment. Domain variety is important as UI conventions differ significantly between website types [6], allowing broader accessibility patterns to be identified.

The experiment runs within a Python-based environment using Playwright, a browser automation library that controls a headless Chromium instance. Playwright loads each website and waits for full rendering before extraction, ensuring that JavaScript-generated content and dynamically applied styles are captured. Computed styles are queried directly from the DOM, reflecting what the user actually sees rather than raw CSS values.

After style extraction, CVD simulation is applied to the extracted colour values. The three simulation types reflect protanopia, deuteranopia, and tritanopia, consistent with the CVD types most commonly addressed in accessibility research [3], [8]. This approach enables controlled evaluation without human participants, addressing the scalability limitations identified in user-based studies [6].

The pipeline follows these sequential stages:

1) **Website Scanning** — Playwright loads each URL and waits for full render.
2) **Component Detection** — UI elements are classified into buttons, links, forms, navigation, headers, and footers.
3) **Style Extraction** — Foreground colour, background colour, and font size are extracted from each component's computed styles.
4) **CVD Simulation** — Colours are mathematically transformed to replicate protanopia, deuteranopia, and tritanopia [3].
5) **WCAG Compliance Evaluation** — Contrast ratios are calculated and evaluated against WCAG 2.2 thresholds: 4.5:1 for normal text and 3:1 for large text [1].
6) **Metric Computation** — Failure rate, contrast loss, and vulnerability scores are derived from the results.
7) **Data Storage** — Results are stored in JSON format for subsequent analysis.

C. Metrics

The primary metric is the WCAG contrast ratio [1]:

$$\text{CR} = \frac{L_{lighter} + 0.05}{L_{darker} + 0.05}$$

Derived metrics, consistent with approaches used in simulation-based accessibility research [3], [4], include:

a) Failure Rate (FR):

$$FR = \frac{\text{Failed Components}}{\text{Total Components}} \tag{1}$$

This provides an overall picture of how many components fail WCAG thresholds before and after CVD simulation.

b) Contrast Loss (CL):

$$CL = CR_{normal} - CR_{CVD} \tag{2}$$

This quantifies the degree of contrast degradation introduced by each simulated CVD condition.

c) Component Vulnerability Score (CVS):

$$CVS = \frac{\text{Components Failing under CVD}}{\text{Total Components of that Type}} \tag{3}$$

This metric enables comparison of vulnerability across component categories, identifying which UI element types are most at risk [6].

D. Method of Analysis

Statistical analysis is performed using IBM SPSS Statistics, selected for its suitability for quantitative, multi-group comparison [6]. Descriptive statistics summarise mean contrast ratios, standard deviations, and pass/fail rates per component category. A one-way ANOVA (Analysis of Variance), a statistical test used to determine whether there are statistically significant differences between the means of three or more independent groups, then tests whether significant differences exist in contrast performance across component types, with post-hoc testing applied where significant results are found. Contrast loss values are compared across the three CVD types to address RQ3, and Component Vulnerability Scores are ranked to identify the most at-risk component categories for RQ2. Results are reported with means, standard deviations, F-values, and significance levels.

E. Validity, Reliability, and Generalisability

Validity is supported by grounding all evaluations in WCAG 2.2 [1] and established CVD simulation models [3], [4], though simulations may not fully reflect the perceptual experience of all individuals with CVD. Reliability is ensured through the fully automated and scripted pipeline, producing consistent results on repeated runs [2]. Generalisability is improved by the varied 20–30 website dataset, though findings are limited to visual contrast and do not cover other accessibility dimensions such as keyboard navigation or screen reader compatibility [1], [6].

F. Ethical Considerations

No personal data is collected and no human participants are involved at any stage. All websites are publicly accessible. Care is taken to respect terms of service and avoid excessive automated requests throughout the evaluation process.

IV. Findings and Results

Findings go here

V. Conclusion

Conclusion goes here

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